Participation in Accessible NLP and ATS as an Empty Signifier: Reflections from a Project on Translating Spoken German into German E2R

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Abstract

Participation has become a near-consensual normative demand in current work on German automatic text simplification (ATS), German Easy-to-Read (E2R, *Leichte Sprache*)[1], and accessible natural language processing (NLP). The current workshop framing explicitly welcomes participatory research involving intended users, while DIN SPEC 33429 formalizes the involvement of people with reading and learning difficulties in producing and reviewing German E2R texts. Against this backdrop, this paper offers a social-scientific reflection from an ongoing interdisciplinary project that aims to develop an application (app) for translating spoken German into German E2R. It argues that participation often functions as an empty signifier: the term attracts broad endorsement but remains operationally underspecified and can refer to divergent practices and distributions of authority. Drawing on practical and conceptual tensions in the project, the paper examines why participation with intended users is difficult to realize and identifies risks including epistemic violence, paternalistic assumptions about cognition, and the normalization of externally defined communicative preferences.

1 Introduction

Research on German E2R translation increasingly combines NLP[2] methods with concerns about usability, human-centered evaluation, and the reduction of barriers in practice. Within German E2R research and practice, participation has become a key normative reference point (Baumann et al., 2025; Schomacker et al., 2024; Zehrer, 2024). This orientation is reinforced by DIN SPEC 33429 (and previous regulations on German E2R), which emphasizes that people with reading and learning difficulties should be involved in the translation and review of German E2R materials and acknowledged as experts in their own right (DIN - Deutsches Institut für Normung e. V., 2025).

The paper starts from a practical discrepancy: although participation has been extensively studied in Disability Studies and adjacent disciplines (Bühler, 2001; Whittaker et al., 2019; Jonsson et al., 2025), the term often remains operationally underspecified in interdisciplinary projects. Here, 'participation' can refer to consultation, testing, review, co-design, or shared decision-making. These practices are not equivalent, yet they are often grouped under the same positive label. The paper develops this argument from an ongoing project that aims to build an app for translating spoken German into German E2R. Given the workshop's technical focus on models, resources, and evaluation pipelines, the paper offers a complementary social-scientific analysis. It is a conceptual and project-based reflection, not a report of a systematic user study.

The central claim is that participation often functions as an "empty signifier" (Laclau, 2007, p. 36). It organizes consensus at the level of discourse while leaving unresolved who defines the problem, who decides what counts as successful translation, and how heterogeneity among intended users is handled. In accessible NLP and ATS, these questions shape training targets, interface decisions, evaluation protocols, and the normative assumptions embedded in the system.

2 Project context: spoken German to German E2R

The project behind this paper aims to develop an app that translates spoken German into German E2R. Such an app sits at the intersection of speech

[1] For more information on terminology see Margot Madina's work on German E2R (Madina et al., 2023)

[2] Although the concept of naturalness applies to language only in a limited sense, the tension is especially marked in German E2R because this variety is deliberately produced rather than emerging spontaneously in social interaction. The established term NLP is nevertheless used here to denote the research field.

processing, German E2R translation, and Disability Studies. Its output must not merely be shorter or more readable in a generic sense; it must be usable for people encountering communicative dis/abling[3] phenomena.[4]

This is precisely where participation becomes both indispensable and difficult. Involving intended users appears necessary if the system is to produce outputs that are useful rather than only technically plausible (Whittaker et al., 2019). At the same time, the project experience suggests that participatory aspirations quickly encounter practical and conceptual limits. Who is considered representative of intended users? At what stage of development should participation occur? What happens when participants differ in their preferences and comprehension? How much authority do participants have over core design choices rather than merely surface-level feedback (Whittaker et al., 2019, p. 22)?

These questions are especially relevant in a research landscape shaped by standardization and evaluation. DIN SPEC 33429 offers an important framework for German E2R practice, but standards cannot dissolve the tensions that arise when heterogeneous user needs are translated into technical requirements (DIN - Deutsches Institut für Normung e. V., 2025). DIN SPEC 33429 identifies people with learning difficulties as the primary target group of German E2R without treating that group as a uniform medical or diagnostic category. It also acknowledges that German E2R may benefit people with diverse communicative needs, including those associated with functional illiteracy, aphasia, dementia, or acquired language and hearing impairments. We therefore treat German E2R as a standardized, deliberately constructed language that cannot account for every individual comprehension profile. This implies that tests with intended users cannot represent every person addressed by E2R communication.

[3]We use the spelling 'dis/ability', which Stock et al. (2025, p. 3, 14) and Wiechern (2025, p. 63) attribute to Goodley (2014, p. ix), because it reflects the reciprocal, non-binary relationship between ability and disability.

[4]Drawing on the onto-epistemological concepts of Karen Barad and Vickie Gauci, we understand dis/ability as a phenomenon co-constituted by intra-acting human, non-human, technological, material, and discursive agencies in material-discursive reconfigurations of the world (Barad, 2007b; Gauci, 2021).

3 Participation as an empty signifier

Following Laclau (2007, p. 36-37), an empty signifier is a term whose meaning remains open enough to unite different actors and demands because it lacks a single fixed meaning. Participation functions as such a signifier insofar as it is widely endorsed while referring to different practices and distributions of authority. Its semantic openness allows researchers, developers, practitioners, and policy actors to agree on the term even when they understand it differently. A project may claim participation when professionals who work with the target group review outputs, when selected target-group members review them, when intended users take part in usability testing, or when they shape the problem definition from the outset. These practices may all be valuable, but they should not be conflated.

In accessible NLP and ATS, the term can become legitimating rather than analytical. Once participation is invoked, projects can present themselves as user-oriented even when important decisions remain centralized in technical or institutional expert cultures. Participation then does not necessarily redistribute epistemic authority; it may simply add legitimacy to decisions taken elsewhere. In this productive but risky sense, the concept stabilizes consensus while obscuring unresolved conflicts over expertise, representation, and design control.

This argument does not dismiss participatory design. It rather calls for more precise participatory claims. In interdisciplinary collaboration among NLP, ATS, and social-scientific research, precision is necessary to connect technical optimization with normative inclusion rather than align them only rhetorically.

Hamraie and Fritsch (2019) argue that the expertise of people with dis/abilities should be acknowledged not only as a matter of social justice but also because accessible technologies participate in forms of world-building. Such world-building should reflect the needs, preferences, and epistemic authority of people with dis/abilities. This requires giving people with dis/abilities power over decision-making in research and development without economic exploitation (Whittaker et al., 2019, p. 18-19, 22) or epistemic violence (Spiel et al., 2020). It also supports an ethics by design approach that integrates normative considerations from the outset (Cavalcante Siebert et al., 2023). The success of meaningful participation by people with

dis/abilities can be studied empirically, as Jonsson et al. (2025) demonstrate, although the perspectives of a few participants cannot straightforwardly stand in for a whole group.

Evaluation is one point at which participatory claims can be made operational: it can examine how intended users influence development and whether the resulting system works in context. Automatic metrics such as BERTScore (Zhang et al., 2019) and LENS (Maddela et al., 2023) assess output quality without capturing how people use a system in practice. Freyer et al. (2027) therefore advocate complementing such metrics with usability studies that examine whether users can act on spoken or written content in realistic settings.[5]

In the account developed by Freyer et al. (2027), usability studies assess systems' efficiency, effectiveness, and users' satisfaction. These dimensions can be examined through formative and summative evaluation (Miesenberger et al., 2020; Freyer et al., 2027). Formative evaluation may include expert interviews during development, whereas summative evaluation assesses a sufficiently developed product with members of the intended user group in the envisaged context of use (Freyer et al., 2027).

Drawing on van der Geest and Velleman (2014), Freyer et al. (2027) argue that efficiency and effectiveness should be assessed by observing how users act on instructions and complete tasks in realistic scenarios. ISO 9241-11 likewise requires usability evaluation to specify the target user group, their objectives, and the contextual factors that shape interaction with the system (DIN - Deutsches Institut für Normung e. V., 2018; Freyer et al., 2027). This approach tests whether ATS systems improve users' ability to act in context, rather than merely whether they understand individual texts or utterances.

4 Tensions in participatory accessible NLP and ATS

A first tension concerns the *practical difficulty* of realizing participation with intended users. In accessible communication projects, participation is often presented as self-evidently desirable, yet time, resources, institutional settings, recruitment constraints, and communicative asymmetries shape how it occurs. These constraints preclude treating participation as a transparent or frictionless design solution.

Our project is developing an app with a non-university social-sector partner to translate complex spoken German into written German E2R in real time, for example, during consultations with doctors. For ethical reasons consistent with an *ethics by design* approach, and for practical reasons such as improving the app's usability, members of the target group should have been involved from the outset in evaluating its design and potential uses. All project partners discussed this objective at the beginning of the project. Nevertheless, challenges have emerged regarding how participation can be implemented and what personnel and time the project can allocate to it.

A second tension concerns *individualization*. Participation is often organized through engagement with individual users or small groups. This is unavoidable to some extent, but in the context of cognitive dis/abilities and related communicative differences, such individualization is methodologically and politically ambivalent. The intended user group is not homogeneous, and differences within it are substantial. If a limited number of participants implicitly stand in for 'the user', heterogeneity is flattened. Participatory design should therefore be complemented by personalization options within the deployed product, so that users can calibrate outputs to their preferences and capacities instead of being forced into a single model of appropriate translation. Personalization can address heterogeneity only partially and cannot substitute for participatory authority.

Our non-university project partner, who serves as a gatekeeper to the target group, has implemented similar projects and established a review structure. This structure consists primarily of individuals who take part in regular exchange meetings, most of which focus on technology-related issues. We therefore assume that these individuals may not encounter some of the difficulties that other members of the target group could experience when using the app. This raises the question of how large the review group should be. The issue is particularly salient in relation to speech and language impairments and learning dis/abilities because the target group is highly heterogeneous. A participatory development process cannot assume that its members have uniform needs or expectations of technological support.

[5] The core principles of usability studies developed by Gould and Lewis (1985) were subsequently systematized for design processes by Nielsen (1992) and standardized in ISO 9241-11 (DIN - Deutsches Institut für Normung e. V., 2018; Bevan et al., 2015), for which a range of practical evaluation instruments has been developed (Wallach and Scholz, 2012).

A third tension concerns epistemic and structural violence (Spiel et al., 2020; Mills et al., 2024, p. 5). Accessible language interventions are often justified by assumptions about what users need in relation to abstraction, complexity, directness, and communicative explicitness. Even when motivated by supportive intentions, such assumptions become epistemically problematic if they define people primarily through deficits or low expectations. What is framed as accessibility may then communicate that little is expected from the addressee. The issue is not whether simplification is needed, but how its rationale is constructed, who authorizes it, and whether it reproduces paternalistic understandings of the people it is intended to support.

In our project, the heterogeneity of the target group became apparent when project discussions repeatedly invoked examples of individual target-group members for whom a proposed testing scenario or app concept might be inappropriate. For instance, when we proposed testing an early click-through prototype with members of the target group, we were told that this would be risky: users might evaluate the app on the basis of the prototype, whose functionality would necessarily be limited, making it difficult to engage them with the fully developed app later. Elsewhere, however, it was emphasized that some members of the target group had very poor short-term memory and that testing processes or appointments scheduled far in advance might be ineffective because users could forget the dates. The arguments pulled in opposite directions but shared an underlying logic: they conceptualized the capacities of the target group through worst-case assumptions, in effect complicating the design and execution of the participatory process. Although this approach may help minimize potential barriers to use, it risks exercising epistemic violence when such assumptions are used to restrict participation.

Such risks are not reducible to individual intentions. They may emerge from institutional arrangements in which welfare-oriented, managerial, scientific, standardizing, and technical rationalities pre-structure who is recognized as a legitimate participant and how participation can occur in practice (Dean, 2010). Such arrangements also shape which contributions count as knowledge (Spiel et al., 2020). This does not imply bad faith on the part of service providers, universities, researchers, developers, or other institutional actors. It identifies structural tensions that participatory accessible technology research should make explicit. A governmentality perspective shifts the analysis from individual attitudes to the rationalities, truth regimes, and practices through which participation is organized (Dean, 2010). It asks how institutional logics define the problem, the user, relevant expertise, and acceptable forms of involvement. In interdisciplinary collaborations, technical, institutional, and disability-political rationalities may apply different criteria. Participation is not a simple corrective to epistemic violence; it can also become a site at which epistemic and structural tensions are reproduced, negotiated, or made visible.

A fourth tension concerns preferences *regarding mitigation and directness* in algorithmic translation. Insult mitigation, for example, may appear to be an obviously desirable feature of a German E2R translation app (Rischer et al., 2026). Yet communicative preferences are not uniform. Some users may prefer softened formulations, while others may want direct renderings in order to understand conflict situations, assess hostility, or decide for themselves what they want to experience. Participation does not resolve this disagreement. Rather, it makes the disagreement visible and requires projects to address it as a design and governance question instead of eliminating it through a technical default.

5 Toward a reflexive participatory framework

Drawing on the project experience discussed above, we propose distinguishing consultation, testing, co-design, and shared decision-making. Projects should document where participation shapes development and where it does not. Such documentation makes participatory claims more precise and credible by specifying whether participation occurs during problem definition, data collection, model and interface design, evaluation, deployment, or revision. These stages permit different distributions of authority.

As a result, participation should be linked to heterogeneity rather than assumed to resolve it. For the app discussed here, personalization is one response to the fact that intended users do not share a single communicative profile. Following Akrich and Barad, projects should diffract the assumptions inscribed in the medium through heterogeneous user experiences (Akrich, 2006; Barad, 2007a). This can help reduce technoableism (Shew, 2020)

and epistemic violence (Spiel et al., 2020). Personalization does not by itself guarantee participatory authority. Therefore, projects must also reflect on the risk of epistemic violence when describing the needs of people with cognitive dis/abilities or related communicative differences. A participatory process that reproduces epistemic violence is not neutral simply because of the formal involvement of users.

6 Practical implications for interdisciplinary projects

Interdisciplinary consortia should negotiate an explicit working definition of participation at the outset instead of assuming that computer scientists, linguists, ethicists, social scientists, and other collaborators mean the same thing. In our project, the term initially functioned as a point of easy agreement that concealed different expectations about the scope of user involvement. A written participation plan should identify the intended participants, the development stages at which they are involved, and the decisions reserved for the technical team. Such a plan is a precondition for ethics by design, not a supplement to it. Without a shared understanding, ethical commitments tend to be deferred to downstream evaluation after consequential design decisions have already been made.

Furthermore, testing arrangements should specify who is an expert in relation to whom. Describing target-group members as experts, as DIN SPEC 33429 does (DIN - Deutsches Institut für Normung e. V., 2025), is an important corrective, but it becomes analytically imprecise if read as a general rather than relational attribution. Both sides bring different expertise: the project team holds expertise in speech processing and system architecture, while participants hold expertise in their communicative practices, the situations in which comprehension breaks down, and what they want to understand. Framing user testing as an *expert evaluation* in this relational sense clarifies what kind of judgment is being solicited. If relatives or support staff speak on behalf of the target group, this substitution should be documented rather than framed as participation by the target group. If recruitment constraints make proxies unavoidable, the method description should state both the substitution and its limitations.

In addition, participation should be integrated into development cycles rather than confined to isolated workshops. In agile settings such as Scrum, this means translating user-facing questions into backlog items and scheduling target-group involvement while sprint results remain revisable. The schedule and format of involvement must also be adapted to participants' communicative and support needs. It is not an option to simply impose the sprint structure on them. Even then, sprint logic bounds participation and does not by itself redistribute authority over the product vision. This arrangement will also surface disagreement between participants, for instance over mitigation and directness, that cannot be resolved by counting preferences. A credible participatory account should document where participants disagreed and how the project resolved the disagreement.

Evaluation should follow the same principle. Usability studies should be scheduled while system components remain revisable and should examine whether users can complete relevant tasks in realistic contexts, not only whether they understand isolated outputs (Freyer et al., 2027).

7 Conclusion

This paper has argued, from the perspective of an ongoing interdisciplinary project on translating spoken German into German E2R, that participation often functions as an empty signifier in accessible technology development. Despite broad support for participation, the concept remains operationally underspecified, refers to different practices, and is difficult to realize with intended users in a substantively strong sense. Participation also entails risks, including individualization, epistemic violence, paternalistic assumptions, and unresolved conflicts over communicative preferences. A reflexive account should make visible where participation remained limited, impossible, or conflictual.

For technology research on accessible NLP and ATS, this is not an external critique but an invitation to sharpen core design concepts. A more reflexive understanding of participation can help interdisciplinary projects move beyond symbolic inclusion toward clearer accounts of authority, heterogeneity, and epistemic responsibility. In that sense, social-scientific reflection is not an alternative to technological progress, but one condition for making technological progress genuinely responsive to the people in whose name it is pursued.

AI acknowledgment

Any use of generative AI in this paper adheres to the ethical guidelines outlined in *Guidelines for Ethical Use and Acknowledgement of Large Language Models in Academic Writing* (Porsdam Mann et al. 2024). Each author has made a substantial contribution, has reviewed the text for accuracy, and assumes responsibility for the integrity of their contributions. *Perplexity Pro* and *DeepL Write* were used to rephrase and stylistically refine individual paragraphs. *DeepL Translator* was used to translate individual phrases.

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